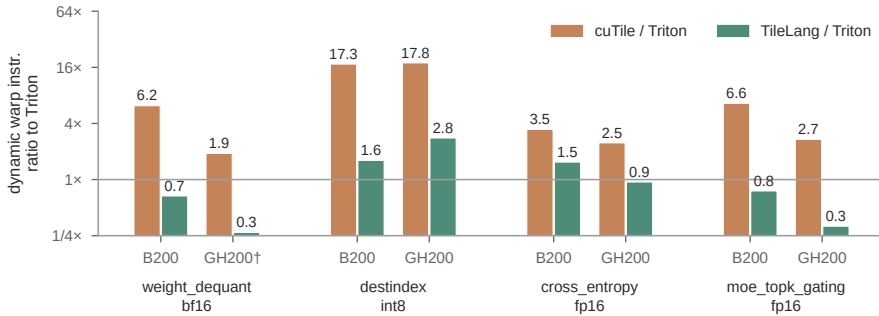


A Indexing and reduction overhead (NVIDIA, smssp_inst_executed.sum, same device)



† GH200 Triton uses a different autotuned config, which inflates its instruction count (confounder, not an architecture effect)

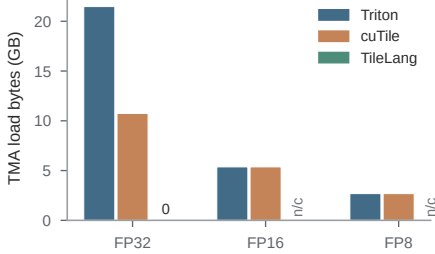
MI300X (Triton, static ISA)

destindex int8:
4 buffer_store_byte per lane,
no vector store (traced)

weight_dequant (control):
constexpr N → multiply-high
index math

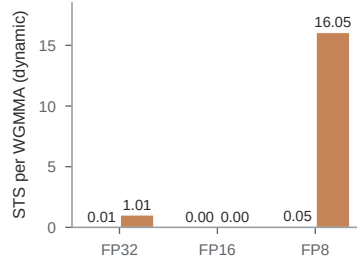
moe_topk_gating (control):
single-wave rows
(DPP + ds_bpermute)

B Matrix operand delivery B200 matmul: TMA bytes

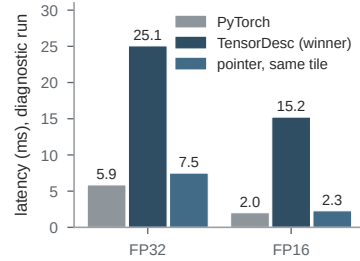


TileLang FP32: legacy HMM via cp.async
(0 B TMA); FP16/FP8 TileLang reports
are reduced collections (n/c)

GH200 matmul: operand re-layout

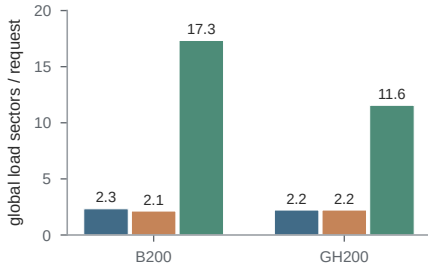


MI300X: descriptor vs pointer

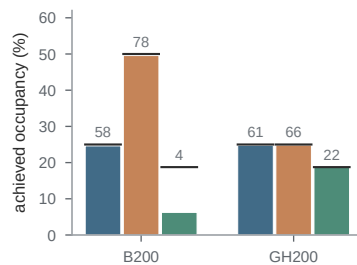


diagnostic run, warmup 2/repeat 10
(not formal). FP32 variant: same
tile and stages, masked tl.store.
FP16 variant also sets num_stages
3 → 2 (not a one-factor change).

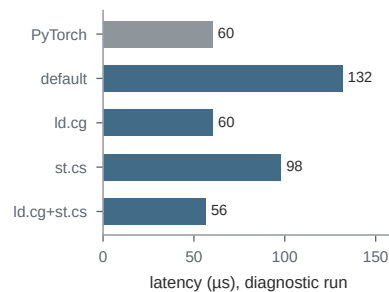
C Gathered staging, latency hiding 1d_conv fp16: load coalescing



bar: achieved, line: theoretical;
number: issue-active %



MI300X vector_add fp32: load cache policy



n = 20,971,520; BLOCK 2048, 4 warps;
warmup 2/repeat 10 (diagnostic);
ld.cg emits nt loads (as ATen);
st.cs has no ISA effect